

jQuery

jQuery Library Inclusion

Exercise 1: Adding jQuery to a Page

- Task: Create a simple HTML page and include jQuery in two ways:

1. Using a CDN link.
2. Using a locally downloaded copy.

- Bonus: Add a

element that says "Hello World!" and log it using jQuery to the console.

Source Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>jQuery CDN Example</title>
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
</head>
<body>

  <h1 id="message">Hello World!</h1>

  <script>
    // Bonus: Wait for the document to fully load, then log the text using jQuery
    $(document).ready(function() {
      var text = $("#message").text();
      console.log("jQuery CDN Log:", text);
    });
  </script>

</body>
</html>
```

Output:

Hello World!

. The \$() Function Exercise 2:

Selecting Elements

- Task: o Create a webpage with: ▪ 1 , 2 , and 1
- o Use \$(document).ready() to ensure the DOM is loaded.
- o Use the \$() function to:
 - Change the text of the
 - Hide one of the elements when the button is clicked

Source Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
```

```

<title>jQuery Selectors Exercise</title>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
<style>
  body { font-family: Arial, sans-serif; margin: 20px; }
  button { margin-top: 10px; padding: 5px 10px; cursor: pointer; }
  .target-p { background-color: #f0f0f0; padding: 5px; }
</style>
</head>
<body>

  <h1 id="main-header">Original Header Text</h1>

  <p class="target-p">Paragraph 1: This paragraph will disappear when you click the button.</p>
  <p>Paragraph 2: This paragraph will stay visible.</p>

  <button id="hide-btn">Click to Hide Paragraph 1</button>

  <script>
    // Ensure the DOM structure is fully loaded before running script
    $(document).ready(function() {

      // 1. Change the text of the <h1> element using ID selector
      $("#main-header").text("The Header Text Has Been Changed!");

      // 2. Hide one specific paragraph when the button is clicked
      $("#hide-btn").click(function() {
        // Selecting by class name to target only the first paragraph
        $(".target-p").hide();
      });

    });
  </script>
</body>
</html>

```

Output:

The Header Text Has Been Changed!

Paragraph 1: This paragraph will disappear when you click the button.

Paragraph 2: This paragraph will stay visible.

Click to Hide Paragraph 1

The Header Text Has Been Changed!

Paragraph 2: This paragraph will stay visible.

Click to Hide Paragraph 1

jQuery Methods

Exercise 3: Use of Common Methods

• Task:

o Add 3 elements with class .box
 o Use the following jQuery methods: ▪ .hide(), .show(), .fadeOut(), .fadeIn(), .toggle() ▪ Apply these methods when clicking respective buttons.

• Bonus: Chain methods like .slideUp().delay(1000).slideDown()

<!DOCTYPE html>

```

<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>jQuery Methods Exercise</title>
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
  <style>
    body { font-family: Arial, sans-serif; margin: 20px; }
    .box-container { display: flex; gap: 15px; margin-bottom: 20px; }

    /* 3 div elements with class .box */
    .box {
      width: 100px;
      height: 100px;
      background-color: #3498db;
      color: white;
      display: flex;
      align-items: center;
      justify-content: center;
      font-weight: bold;
      border-radius: 8px;
    }

    .controls { display: flex; gap: 8px; flex-wrap: wrap; }
    button { padding: 8px 12px; font-weight: bold; cursor: pointer; }
    .bonus-btn { background-color: #2ecc71; color: white; border: none; border-radius: 4px; }
  </style>
</head>
<body>

  <div class="box-container">
    <div class="box">Box 1</div>
    <div class="box">Box 2</div>
    <div class="box">Box 3</div>
  </div>

  <div class="controls">
    <button id="btn-hide">.hide()</button>
    <button id="btn-show">.show()</button>
    <button id="btn-fadeout">.fadeOut()</button>
    <button id="btn-fadein">.fadeIn()</button>
    <button id="btn-toggle">.toggle()</button>

    <button id="btn-bonus" class="bonus-btn">Trigger Chain (.slideUp...)</button>
  </div>

  <script>
    $(document).ready(function() {

      // 1. Hide boxes instantly
      $("#btn-hide").click(function() {
        $(".box").hide();
      });
    });
  </script>

```

```

});

// 2. Show boxes instantly
$("#btn-show").click(function() {
    $(".box").show();
});

// 3. Smoothly fade out boxes over 800ms
$("#btn-fadeout").click(function() {
    $(".box").fadeOut(800);
});

// 4. Smoothly fade back in over 800ms
$("#btn-fadein").click(function() {
    $(".box").fadeIn(800);
});

// 5. Toggle visibility status (alternates hide/show)
$("#btn-toggle").click(function() {
    $(".box").toggle(400);
});

// Bonus: Method Chaining
// Slides the boxes away, waits for 1 second (1000ms), then slides them back down
$("#btn-bonus").click(function() {
    $(".box").slideUp(500).delay(1000).slideDown(500);
});

});
</script>
</body>
</html>

```



Working with Events

Exercise 5: Handling User Interactions

- Task: o Create a page with: ▪ A with id #colorBtn ▪ A with id #colorBox
- o Use jQuery to: ▪ Change the background color of #colorBox to red when the button is clicked.
- Change it back to white when double-clicked.

Source Code:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">

```

```
<title>jQuery Events Exercise</title>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
<style>
  body {
    font-family: Arial, sans-serif;
    margin: 20px;
    display: flex;
    flex-direction: column;
    gap: 15px;
    align-items: flex-start;
  }

  /* A div with id #colorBox */
  #colorBox {
    width: 200px;
    height: 200px;
    border: 2px solid #333;
    background-color: white; /* Initial color */
    transition: background-color 0.3s ease; /* Smooth transition */
    border-radius: 8px;
  }

  /* A button with id #colorBtn */
  #colorBtn {
    padding: 10px 15px;
    font-weight: bold;
    cursor: pointer;
    background-color: #34495e;
    color: white;
    border: none;
    border-radius: 4px;
  }

  #colorBtn:hover {
    background-color: #2c3e50;
  }
</style>
</head>
<body>

  <button id="colorBtn">Click / Double-Click Me</button>
  <div id="colorBox"></div>

<script>
  $(document).ready(function() {

    // 1. Single click event: Change background color to red
    $("#colorBtn").click(function() {
      $("#colorBox").css("background-color", "red");
    });
  });
</script>
```

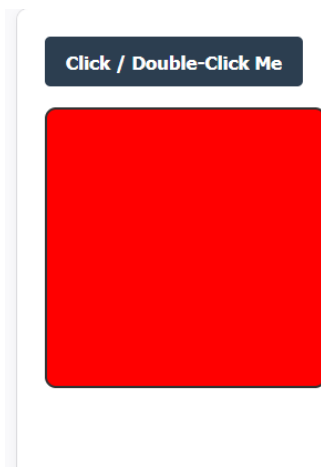
```

        // 2. Double click event: Change background color back to white
        $("#colorBtn").dblclick(function() {
            $("#colorBox").css("background-color", "white");
        });

    });
</script>

</body>
</html>

```



Event Helpers

Exercise 6: Mouse and Keyboard Events

- Task:
 - o Create a paragraph with some text.
 - o Use the following event helpers: ▪ .click() ▪ .dblclick() ▪ .mouseenter() and .mouseleave() ▪ .keypress() on an input field
 - o Make each event do something visible (e.g., change background color, add text, show an alert, etc.)

Source Code:

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>jQuery Event Helpers Exercise</title>
    <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
    <style>
        body { font-family: Arial, sans-serif; margin: 30px; line-height: 1.6; }

        /* Stylized paragraph box */
        #interactive-text {
            padding: 20px;

```

```

        background-color: #f5f6fa;
        border: 2px dashed #7f8c8d;
        border-radius: 8px;
        font-size: 18px;
        text-align: center;
        cursor: pointer;
        user-select: none; /* Prevents text highlighting on double click */
        transition: all 0.2s ease;
        margin-bottom: 20px;
    }

    .input-group { margin-top: 20px; }
    input { padding: 8px; width: 250px; font-size: 14px; }
    #key-log { color: #e74c3c; font-weight: bold; margin-left: 10px; }
</style>
</head>
<body>

    <p id="interactive-text">
        Hover over me, click me, or double-click me!
    </p>

    <div class="input-group">
        <label for="text-input">Type something here: </label>
        <input type="text" id="text-input" placeholder="Press keys to trigger .keypress()...">
        <span id="key-log"></span>
    </div>

    <script>
        $(document).ready(function() {
            var $textBox = $("#interactive-text");

            // 1. .mouseenter() - Triggers when mouse hovers inside
            $textBox.mouseenter(function() {
                $(this).css({
                    "background-color": "#ffeaa7",
                    "border-color": "#fdbc6e"
                }).text("Mouse entered the zone!");
            });

            // 2. .mouseleave() - Triggers when mouse moves away
            $textBox.mouseleave(function() {
                $(this).css({
                    "background-color": "#f5f6fa",
                    "border-color": "#7f8c8d"
                }).text("Hover over me, click me, or double-click me!");
            });

            // 3. .click() - Triggers on a single left-click
            $textBox.click(function() {
                $(this).css("background-color": "#9b59b6")
            });
        });
    </script>

```

```
        .text(" 🌟 Single Click Detected! 🌟 ");
    });

    // 4. .dblclick() - Triggers on a rapid double-click
    $textBox.dblclick(function() {
        $(this).css("background-color": "#2ecc71")
        .text(" 🎉 Double Click Registered! 🎉 ");
    });

    // 5. .keypress() - Triggers when a character key is pressed inside the input
    $("#text-input").keypress(function(event) {
        // event.key gets the actual letter or symbol typed
        $("#key-log").text("Last key pressed: " + event.key + "").fadeIn(100).fadeOut(800);
    });

    });
</script>

</body>
</html>
```

Output:

Hover over me, click me, or double-click me!

Type something here: